NeuralMetricsWeb — EEG Analysis Methods

A guide for drafting the Methods section of a manuscript

How to use this guide. The text below describes the data-collection, artifact-handling, preprocessing and spectral methods implemented in the NeuralMetricsWeb Python pipeline, written so it can be adapted directly into a manuscript Methods section. Passages shaded and marked **[TO BE COMPLETED BY THE AUTHOR]** are study-specific and must be filled in. Numerical parameters reflect the current pipeline configuration; confirm they match the software version used for your study (record the version/commit).

1. EEG acquisition and data import

Perioperative frontal EEG was acquired with **[TO BE COMPLETED BY THE AUTHOR**: recording monitor and model] and exported as European Data Format files (EDF/EDF+) for Masimo SedLine, or as the vendor raw export with its 1-Hz trend (.spa) and event-log (.e_a) sidecars for Medtronic BIS Vista. All analyses ran on the local device; no recording was uploaded. The analysis channel was selected automatically by electrode label with a frontal-montage preference. **[TO BE COMPLETED BY THE AUTHOR**: number of recordings/subjects, anaesthetic technique, sampling rate (Hz), mains frequency (the line-noise notch is configured at 60 Hz), and the ethics-approval and informed-consent statement.]

2. Electrode montage (common practice)

Processed-EEG depth-of-anaesthesia monitoring uses a frontal montage. The Masimo SedLine sensor is a self-adhesive bilateral array of six electrodes placed across the forehead, providing left and right frontal derivations (channels L1/L2 and R1/R2) at approximately Fp1 and Fp2 with lateral frontal positions near F7 and F8, a reference near the midline forehead (≈ Fpz) and a ground (≈ Fz). The Medtronic BIS sensor (e.g., BIS Quatro) is a four-electrode frontal strip placed across the forehead at approximately Fp1/Fpz with a lateral fronto-temporal position, referenced within the sensor. A single frontal derivation was used for analysis. **[TO BE COMPLETED BY THE AUTHOR**: state the exact sensor, electrode positions and the analysis channel used.]

3. Session assembly

When a case comprised multiple files, files were ordered by their header start time and grouped into a single recording when the inter-file gap was ≤ 30 min and any overlap ≤ 5 s. Within a recording, consecutive files were concatenated into one continuous block when the gap between them was ≤ 1 s (overlapping samples discarded); larger gaps began a new block, preserving true recording gaps on the session clock. A single dominant sampling rate was enforced and any off-rate file excluded. Sensor-off time was removed by a flat-signal test (1-s windows; a window was live when its standard deviation exceeded 0.5 µV): blocks with fewer than 50 % live windows were dropped, and leading and trailing flat regions were trimmed.

4. Artifact handling and signal-quality control

Quality control operated on 4-s windows with 50 % overlap. A window was rejected if it failed any of four gates: (i) *flat* — standard deviation < 0.5 µV; (ii) *saturation* — any sample within half a quantization step of the channel's physical minimum or maximum; (iii) *stuck* — an identical raw value sustained for ≥ 1 s; and (iv) *amplitude* — any sample exceeding 150 µV. Only windows passing all active gates contributed to spectral estimation. Overall signal composition was

summarized as three ratios over all windows that sum to 100 %: the clipping ratio (eCR; saturation or over-amplitude), the flat-line ratio (eFR; flat or stuck) and the true-EEG ratio (eTR).

Additional export-quality checks flagged clipping percentage, amplitude quantization and effective bit depth, within-recording resolution changes, feed/sweep-speed changes, and evidence of monitor pre-filtering. Periodic device impedance checks — isolated low-amplitude segments of ≤ 10 s (sub-gaps bridged over ≤ 4 s) separated from genuine suppression by ≥ 120 s of analysed EEG on both sides — were classified as indeterminate rather than cerebral activity and excluded from burst-suppression quantification.

5. Preprocessing

Each continuous block was filtered before segmentation with a zero-phase, 4th-order Butterworth band-pass (0.5–40 Hz), implemented as second-order sections and applied forward and backward (scipy.signal.sosfiltfilt); the upper edge was clamped to 0.95 × Nyquist for low-sampling-rate recordings. Line noise was removed with a zero-phase IIR notch at 60 Hz (quality factor 30). Blocks too short to filter were de-meaned only.

6. Spectral analysis (Welch's method)

Power spectral density (PSD) was estimated by Welch's method. The filtered signal was divided into 4-s segments with 50 % overlap (2-s hop); each segment was de-meaned and tapered with a Hann window. The one-sided periodogram was computed via the real FFT, with the squared magnitude normalized by $f_s \cdot \Sigma(w^2)$ and interior bins doubled, giving a frequency resolution of 0.25 Hz. The PSD was averaged across all QC-valid segments pooled over the continuous blocks. Absolute band power (μV^2) was obtained by integrating the averaged PSD over delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz) and total (0.5–30 Hz); relative band power was each band's share of total 0.5–30 Hz power. The spectral edge frequency (SEF95) was the lowest frequency below which 95 % of total power fell. The density spectral array (DSA) was the per-30-s median PSD of QC-valid windows on a shared timeline.

7. Burst-suppression analysis

Suppression was detected on the filtered signal with a line-length-and-amplitude criterion: a sample was a suppression candidate when its centred line length (0.2807-s window) remained ≤ 40 µV and its amplitude ≤ 5 µV; candidate episodes shorter than 1 s were discarded. Saturated, stuck and impedance-check samples were treated as indeterminate and excluded. The burst-suppression ratio (BSR) was the percentage of determinate EEG classified as suppression, reported for the analysed window and as a per-minute trend.

8. Composite index

A processed-EEG composite index was computed from the frontal signal using an open reimplementation of the depth-of-anaesthesia algorithm, with the signal resampled to 128 Hz by polyphase filtering prior to the composite computation.

9. Implementation and reproducibility

The pipeline was implemented in Python using NumPy and SciPy and executed deterministically in the browser via Pyodide, so a given recording and configuration always yield identical results. **[TO BE COMPLETED BY THE AUTHOR**: software name and version/commit, and the date of

analysis.]

References

1. Kemp B, Värri A, Rosa AC, Nielsen KD, Gade J. A simple format for exchange of digitized polygraphic recordings. *Electroencephalogr Clin Neurophysiol.* 1992;82(5):391–393. doi:10.1016/0013-4694(92)90009-7

2. Kemp B, Oliván J. European data format 'plus' (EDF+), an EDF alike standard format for the exchange of physiological data. *Clin Neurophysiol.* 2003;114(9):1755–1761. doi:10.1016/S1388-2457(03)00123-8

3. Welch PD. The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Trans Audio Electroacoust.* 1967;15(2):70–73. doi:10.1109/TAU.1967.1161901

4. Rampil IJ. A primer for EEG signal processing in anesthesia. *Anesthesiology.* 1998;89(4):980–1002. doi:10.1097/00000542-199810000-00023

5. Purdon PL, Sampson A, Pavone KJ, Brown EN. Clinical electroencephalography for anesthesiologists: Part I. *Anesthesiology.* 2015;123(4):937–960. doi:10.1097/ALN.0000000000000841

6. Särkelä M, Mustola S, Seppänen T, et al. Automatic analysis and monitoring of burst suppression in anesthesia. *J Clin Monit Comput.* 2002;17(2):125–134. doi:10.1023/A:1016393904439

7. Bennett WR. Spectra of quantized signals. *Bell Syst Tech J.* 1948;27(3):446–472. doi:10.1002/j.1538-7305.1948.tb01340.x

8. von Dincklage F, et al. *J Clin Monit Comput.* 2021;35(5):1047–1054. doi:10.1007/s10877-020-00578-9 [confirm article title]

9. Bøgli SY, et al. *Physiol Meas.* 2024;45(5):05NT02. doi:10.1088/1361-6579/ad46e4 [confirm article title]

10. Connor CW. Open reimplementation of the BIS algorithms for depth of anesthesia. *Anesth Analg.* 2022;135(4):855–864. doi:10.1213/ANE.0000000000006119

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